Alderbrook Insurance Group: Modernizing Claims Intake with AI-Assisted Triage

Case study · Data & AI practice · Insurance

Client	Alderbrook Insurance Group
Industry	Insurance
Service line	Data & AI
Engagement type	Claims Processing Transformation
Company size	Mid-size regional insurer, ~2,100 employees
Project scale	National claims operations, 6 regional processing centers
Client relationship	Existing strategic client, engaged for a second phase of work
Relationship duration	4 years

BUSINESS PROBLEM

Alderbrook processed roughly 40,000 property and casualty claims a month through a largely manual, paper-and-email intake pipeline. Adjusters spent the majority of their time re-keying data from scanned documents rather than evaluating claims, average processing time had crept past nine days, and a growing backlog was driving customer complaints and adjuster attrition. Leadership needed a way to triage and route claims automatically without lowering the accuracy of fraud detection.

SOLUTION

We built an AI-assisted claims intake platform that ingests scanned forms, photos, and adjuster notes, extracts structured data automatically, and routes each claim to the right queue based on complexity and fraud risk. Adjusters review a pre-filled claim instead of transcribing one from scratch, and a machine learning model flags anomalous claims for manual investigation before payout.

TECHNOLOGY STACK

Cloud document processing and OCR, a Python-based data pipeline, a machine learning fraud-scoring model, and a workflow engine integrated with Alderbrook's existing claims management system.

ARCHITECTURE

Event-driven ingestion pipeline with automated document extraction, a real-time risk-scoring service, and a rules-and-ML hybrid routing layer that assigns each claim to the appropriate adjuster queue.

OUTCOME

9 days → 2.5 days average claims processing time	**63%** reduction in manual data entry	**+18%** fraud flags caught pre-payout

Average claims processing time dropped from nine days to under three, freeing adjusters to focus on complex cases and customer communication rather than data entry. Fraud detection improved despite the faster cycle time, since the risk-scoring model runs automatically on every claim rather than relying solely on adjuster judgment.

FINANCIAL IMPACT

Reduced claims-handling labor costs and a measurable drop in fraudulent payouts more than offset the cost of the engagement within the first year.

INVESTMENT

Mid-six-figure engagement delivered over five months, followed by an ongoing managed-service arrangement for model monitoring and tuning.